

# 15

## Linear Quadratic Optimization

This chapter considers time domain characterizations of Hankel operators and Toeplitz operators by means of some related quadratic optimizations. These characterizations will be used to prove a max-min problem which is the key to the  $\mathcal{H}_\infty$  theory considered in the next chapter.

### 15.1 Hankel Operators

Let  $G(s)$  be a stable real rational transfer matrix with a state space realization

$$\begin{aligned}\dot{x} &= Ax + Bw \\ z &= Cx + Dw.\end{aligned}\tag{15.1}$$

Consider first the problem of using an input  $w \in \mathcal{L}_{2-}$  to maximize  $\|P_+ z\|_2^2$ . This is exactly the standard problem of computing the Hankel norm of  $G$ , i.e., the induced norm of the Hankel operator

$$P_+ M_G : \mathcal{H}_2^\perp \rightarrow \mathcal{H}_2,$$

and the norm can be expressed in terms of the controllability Gramian  $L_c$  and observability Gramian  $L_o$ :

$$AL_c + L_c A^* + BB^* = 0 \quad A^* L_o + L_o A + C^* C = 0.$$

Although this result is well-known, we will include a time-domain proof similar in technique to the proofs of the optimal  $\mathcal{H}_2$  and  $\mathcal{H}_\infty$  control results.

**Lemma 15.1**  $\inf_{w \in \mathcal{L}_{2-}} \{ \|w\|_2^2 \mid x(0) = x_0 \} = x_0^* y_0$  where  $y_0$  solves  $L_c y_0 = x_0$ .

**Proof.** Assume  $(A, B)$  is controllable; otherwise, factor out the uncontrollable subspace. Then  $L_c$  is invertible and  $y_0 = L_c^{-1} x_0$ . Moreover,  $w \in \mathcal{L}_{2-}$  can be used to produce any  $x(0) = x_0$  given  $x(-\infty) = 0$ . We need to show

$$\inf_{w \in \mathcal{L}_{2-}} \{ \|w\|_2^2 \mid x(0) = x_0 \} = x_0^* L_c^{-1} x_0. \quad (15.2)$$

To show this, we can differentiate  $x(t)^* L_c^{-1} x(t)$  along the solution of (15.1) for any given input  $w$  as follows:

$$\frac{d}{dt}(x^* L_c^{-1} x) = \dot{x}^* L_c^{-1} x + x^* L_c^{-1} \dot{x} = x^* (A^* L_c^{-1} + L_c^{-1} A) x + 2\langle w, B^* L_c^{-1} x \rangle.$$

Using  $L_c$  equation to substitute for  $A^* L_c^{-1} + L_c^{-1} A$  and completion of the squares gives

$$\frac{d}{dt}(x^* L_c^{-1} x) = \|w\|^2 - \|w - B^* L_c^{-1} x\|^2.$$

Integration from  $t = -\infty$  to  $t = 0$  with  $x(-\infty) = 0$  and  $x(0) = x_0$  gives

$$x_0^* L_c^{-1} x_0 = \|w\|_2^2 - \|w - B^* L_c^{-1} x\|_2^2 \leq \|w\|_2^2.$$

If  $w(t) = B^* e^{-A^* t} L_c^{-1} x_0 = B^* L_c^{-1} e^{(A + B B^* L_c^{-1})t} x_0$  on  $(-\infty, 0]$ , then  $w \in \mathcal{L}_{2-}$ ,  $w = B^* L_c^{-1} x$  and equality is achieved, thus proving (15.2).  $\square$

**Lemma 15.2**  $\sup_{w \in \mathcal{B}\mathcal{L}_{2-}} \|P_+ z\|_2^2 = \sup_{w \in \mathcal{B}\mathcal{H}_2^\perp} \|P_+ M_G w\|_2^2 = \rho(L_o L_c).$

**Proof.** Given  $x(0) = x_0$  and  $w = 0$ , for  $t \geq 0$  the norm of  $z(t) = C e^{A t} x_0$  can be found from

$$\|P_+ z\|_2^2 = \int_0^\infty x_0^* e^{A^* t} C^* C e^{A t} x_0 dt = x_0^* L_o x_0.$$

Combine this result with Lemma 15.1 to give

$$\sup_{w \in \mathcal{B}\mathcal{L}_{2-}} \|P_+ z\|_2^2 = \sup_{0 \neq w \in \mathcal{L}_{2-}} \frac{\|P_+ z\|_2^2}{\|w\|_2^2} = \max_{x_0 \neq 0} \frac{x_0^* L_o x_0}{x_0^* L_c^{-1} x_0} = \rho(L_o L_c).$$

$\square$

**Remark 15.1** Another useful way to characterize the Hankel norm is to examine the following quadratic optimization with initial condition  $x(-\infty) = 0$ :

$$\sup_{w \in \mathcal{L}_{2-}} \{ \|P_+ z\|_2^2 - \beta^2 \|w\|_2^2 \}.$$

It is easy to see from the definition of the Hankel norm that

$$\sup_{0 \neq w \in \mathcal{L}_{2-}} \frac{\|P_+ z\|_2}{\|w\|_2} \leq \beta$$

iff

$$\sup_{0 \neq w \in \mathcal{L}_{2-}} \{ \|P_+ z\|_2^2 - \beta^2 \|w\|_2^2 \} \leq 0.$$

So the Hankel norm is equal to the smallest  $\beta$  such that the above inequality holds. Now

$$\begin{aligned} \sup_{w \in \mathcal{L}_{2-}} \{ \|P_+ z\|_2^2 - \beta^2 \|w\|_2^2 \} &= \sup_{x_0 \in \mathbb{R}^n} (x_0^* L_o x_0 - \beta^2 x_0^* L_c^{-1} x_0) \\ &= \begin{cases} 0, & \rho(L_o L_c) \leq \beta^2; \\ +\infty, & \rho(L_o L_c) > \beta^2. \end{cases} \end{aligned}$$

Hence the Hankel norm is equal to the square root of  $\rho(L_o L_c)$ . ♥

## 15.2 Toeplitz Operators

If a transfer matrix  $G \in \mathcal{RH}_\infty$  and  $\|G\|_\infty < 1$ , then by Corollary 13.24, the Hamiltonian matrix

$$H = \begin{bmatrix} A + BR^{-1}D^*C & BR^{-1}B^* \\ -C^*(I + DR^{-1}D^*)C & -(A + BR^{-1}D^*C)^* \end{bmatrix}, \quad R = \gamma^2 I - D^*D$$

with  $\gamma = 1$  is in  $\text{dom}(\text{Ric})$ ,  $X = \text{Ric}(H) \geq 0$ ,  $A + BR^{-1}(B^*X + D^*C)$  is stable and

$$A^*X + XA + (XB + C^*D)R^{-1}(B^*X + D^*C) + C^*C = 0. \quad (15.3)$$

The following lemma offers yet another consequence of  $\|G\|_\infty < 1$ . (Recall that the  $\mathcal{H}_\infty$  norm of a stable matrix is the Toeplitz operator norm.)

**Lemma 15.3** *Suppose  $\|G\|_\infty < 1$  and  $x(0) = x_0$ . Then*

$$\sup_{w \in \mathcal{L}_{2+}} (\|z\|_2^2 - \|w\|_2^2) = x_0^* X x_0$$

*and the sup is achieved.*

**Proof.** We can differentiate  $x(t)^*Xx(t)$  as in the last section, use the Riccati equation (15.3) to substitute for  $A^*X + XA$ , and complete the squares to get

$$\frac{d}{dt}(x^*Xx) = -\|z\|^2 + \|w\|^2 - \|R^{-1/2}[Rw - (B^*X + D^*C)x]\|^2.$$

If  $w \in \mathcal{L}_{2+}$ , then  $x \in \mathcal{L}_{2+}$ , so integrating from  $t = 0$  to  $t = \infty$  gives

$$\|z\|_2^2 - \|w\|_2^2 = x_0^*Xx_0 - \|R^{-1/2}[Rw - (B^*X + D^*C)x]\|_2^2 \leq x_0^*Xx_0. \quad (15.4)$$

If we let  $w = R^{-1}(B^*X + D^*C)x = R^{-1}(B^*X + D^*C)e^{[A+BR^{-1}(B^*X+D^*C)]t}x_0$ , then  $w \in \mathcal{L}_{2+}$  because  $A + BR^{-1}(B^*X + D^*C)$  is stable. Thus the inequality in (15.4) can be made an equality and the proof is complete. Note that the sup is achieved for a  $w$  which is a linear function of the state.  $\square$

As a direct consequence of this lemma, we have the following

**Corollary 15.4** *Let  $x(0) = x_0$  and  $X_0 = X_0^* > 0$ .*

(i) *if  $\|G\|_\infty < \gamma$  and  $X = Ric(H)$ . Then*

$$\begin{aligned} & \sup_{0 \neq (x_0, w) \in \mathbb{R}^n \times \mathcal{L}_2[0, \infty)} \{ \|z\|_2^2 - \gamma^2(\|w\|_2^2 + x_0^*X_0x_0) \} \\ &= \sup_{0 \neq x_0 \in \mathbb{R}^n} \{ x_0^*Xx_0 - \gamma^2 x_0^*X_0x_0 \} \\ & \begin{cases} < 0, & \lambda_{max}(X - \gamma^2 X_0) < 0; \\ = 0, & \lambda_{max}(X - \gamma^2 X_0) = 0; \\ = +\infty, & \lambda_{max}(X - \gamma^2 X_0) > 0. \end{cases} \end{aligned}$$

(ii)  $\sup_{0 \neq (x_0, w) \in \mathbb{R}^n \times \mathcal{L}_2[0, \infty)} \frac{\|P_+ z\|_2^2}{\|w\|_2^2 + x_0^*X_0x_0} < \gamma^2$  if and only if  $\bar{\sigma}(D) < \gamma$ ,  $H \in dom(Ric)$ , and  $\lambda_{max}(X - \gamma^2 X_0) < 0$ .

**Remark 15.2** The matrix  $X_0$  has the interpretation of the confidence on the initial condition  $x_0$ . So if  $\underline{\sigma}(X_0)$  is small, then the initial condition is probably not known very well. In that case  $\gamma_{min}$  will be large where  $\gamma_{min}$  denotes the smallest  $\gamma$  such that  $\lambda_{max}(X - \gamma^2 X_0) \leq 0$ . On the other hand, a large  $\underline{\sigma}(X_0)$  implies that the initial condition is known very well and that  $\gamma_{min}$  is determined essentially by the condition  $H \in dom(Ric)$ .  $\heartsuit$

A dual version of Lemma 15.3 can also be obtained and is useful in characterizing the so-called  $2 \times 2$  block mixed Hankel-Toeplitz operator. To do that, we first note that<sup>1</sup>

$$G^T(s) = \left[ \begin{array}{c|c} A^* & C^* \\ \hline B^* & D^* \end{array} \right]$$

<sup>1</sup>Note that since the system matrices are real,  $A^T = A^*$ ,  $B^T = B^*$ , etc. The conjugate transpose is used here for the transpose for the sake of consistency in notation.

and  $\|G\|_\infty < 1$  iff  $\|G^T\|_\infty < 1$ . Let  $J$  denote the following Hamiltonian matrix

$$J = \begin{bmatrix} A^* & 0 \\ -BB^* & -A \end{bmatrix} + \begin{bmatrix} C^* \\ -BD^* \end{bmatrix} \tilde{R}^{-1} \begin{bmatrix} DB^* & C \end{bmatrix}$$

where  $\tilde{R} := I - DD^*$ . Then  $J \in \text{dom}(\text{Ric})$ ,  $Y = \text{Ric}(J) \geq 0$ ,  $A + (YC^* + BD^*)\tilde{R}^{-1}C$  is stable and

$$AY + YA^* + (YC^* + BD^*)\tilde{R}^{-1}(CY + DB^*) + BB^* = 0 \quad (15.5)$$

if  $\|G\|_\infty < 1$ . For simplicity, we shall assume that  $(A, B)$  is controllable; hence,  $Y > 0$ . The case in which  $Y$  is singular can be obtained by restricting  $x_0 \in \text{Im}(Y)$  and replacing  $Y^{-1}$  by  $Y^+$ .

**Lemma 15.5** *Suppose  $\|G\|_\infty < 1$  and  $(A, B)$  is controllable. Then*

$$\sup_{w \in \mathcal{L}_{2-}} \{ \|P_- z\|_2^2 - \|w\|_2^2 \mid x(0) = x_0 \} = -x_0^* Y^{-1} x_0$$

and the sup is achieved.

**Proof.** Analogous to the proof of Lemma 15.3, we can differentiate  $x(t)^* Y^{-1} x(t)$ , use the Riccati equation (15.5) to substitute for  $AY + YA^*$ , and complete the squares to get

$$\frac{d}{dt}(x^* Y^{-1} x) = -\|z\|^2 + \|w\|^2 - \|R^{-1/2}[Rw - (B^* Y^{-1} + D^* C)x]\|^2$$

where  $R = I - D^* D > 0$ . If  $w \in \mathcal{L}_{2-}$ , then  $x \in \mathcal{L}_{2-}$  and  $x(-\infty) = 0$ ; so integrating from  $t = -\infty$  to  $t = 0$  gives

$$\|z\|_2^2 - \|w\|_2^2 = -x_0^* Y^{-1} x_0 - \|R^{-1/2}[Rw - (B^* Y^{-1} + D^* C)x]\|_2^2 \leq -x_0^* Y^{-1} x_0. \quad (15.6)$$

If we let  $w = R^{-1}(B^* Y^{-1} + D^* C)x = R^{-1}(B^* Y^{-1} + D^* C)e^{[A + BR^{-1}(B^* Y^{-1} + D^* C)]t} x_0$ , then  $w \in \mathcal{L}_{2-}$  because  $A + BR^{-1}(B^* Y^{-1} + D^* C) = -Y\{A + (YC^* + BD^*)\tilde{R}^{-1}C\}^* Y^{-1}$  and  $A + (YC^* + BD^*)\tilde{R}^{-1}C$  is stable. Thus the inequality can be made an equality and the proof is complete.  $\square$

### 15.3 Mixed Hankel-Toeplitz Operators

Now suppose that the input is partitioned so that  $B = \begin{bmatrix} B_1 & B_2 \end{bmatrix}$ ,  $D = \begin{bmatrix} D_1 & D_2 \end{bmatrix}$ ,

$$G(s) = \begin{bmatrix} G_1(s) & G_2(s) \end{bmatrix} \in \mathcal{RH}_\infty,$$

and  $w$  is partitioned conformably. Then  $\|G_2\|_\infty < 1$  iff  $\bar{\sigma}(D_2) < 1$ , and

$$H_W := \begin{bmatrix} A & 0 \\ -C^*C & -A^* \end{bmatrix} + \begin{bmatrix} B_2 \\ -C^*D_2 \end{bmatrix} R_2^{-1} \begin{bmatrix} D_2^*C & B_2^* \end{bmatrix}$$

is in  $\text{dom}(\text{Ric})$  where  $R_2 := I - D_2^*D_2 > 0$ . For  $H_W \in \text{dom}(\text{Ric})$ , define  $W = \text{Ric}(H_W)$ . Let

$$w \in \mathcal{W} := \mathcal{H}_2^\perp \oplus \mathcal{L}_2 = \left\{ \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \mid w_1 \in \mathcal{H}_2^\perp, w_2 \in \mathcal{L}_2 \right\}. \quad (15.7)$$

We are interested in a test for  $\sup_{w \in \mathcal{BW}} \|P_+z\|_2 < 1$ , or, equivalently,

$$\sup_{w \in \mathcal{BW}} \|\Gamma w\|_2 < 1 \quad (15.8)$$

where  $\Gamma = P_+[M_{G_1} \ M_{G_2}] : \mathcal{W} \rightarrow \mathcal{H}_2$  is a mixed Hankel-Toeplitz operator:

$$\begin{aligned} \Gamma \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} &= P_+ \begin{bmatrix} G_1 & G_2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \quad w_1 \in \mathcal{H}_2^\perp, \ w_2 \in \mathcal{L}_2 \\ &= P_+ \begin{bmatrix} G_1 & G_2 \end{bmatrix} P_-w + P_+G_2P_+w_2. \end{aligned}$$

Thus  $\Gamma$  is the sum of the Hankel operator  $P_+M_G : \mathcal{H}_2^\perp \oplus \mathcal{H}_2^\perp \rightarrow \mathcal{H}_2$  and the Toeplitz operator  $P_+M_{G_2} : \mathcal{H}_2 \rightarrow \mathcal{H}_2$ . The following lemma generalizes Corollary 13.24 ( $B_1 = 0, D_1 = 0$ ) and Lemma 15.2 ( $B_2 = 0, D_2 = 0$ ).

**Lemma 15.6**  $\sup_{w \in \mathcal{BW}} \|\Gamma w\|_2 < 1$  iff the following two conditions hold:

- (i)  $\bar{\sigma}(D_2) < 1$  and  $H_W \in \text{dom}(\text{Ric})$ ;
- (ii)  $\rho(WL_c) < 1$ .

**Proof.** By Corollary 13.24, condition (i) is necessary for (15.8), so we will prove that given condition (i), (15.8) holds iff condition (ii) holds. We will do this by showing, equivalently, that  $\rho(WL_c) \geq 1$  iff  $\sup_{w \in \mathcal{BW}} \|\Gamma w\|_2 \geq 1$ . By definition of  $\mathcal{W}$ , if  $w \in \mathcal{W}$  then

$$\|P_+z\|_2^2 - \|w\|_2^2 = \|P_+z\|_2^2 - \|P_+w_2\|_2^2 - \|P_-w\|_2^2.$$

Note that the last term only contributes to  $\|P_+z\|_2^2$  through  $x(0)$ . Thus if  $L_c$  is invertible, then Lemma 15.1 and 15.3 yield

$$\sup_{w \in \mathcal{W}} \{ \|P_+z\|_2^2 - \|w\|_2^2 \mid x(0) = x_0 \} = x_0^* W x_0 - x_0^* L_c^{-1} x_0 \quad (15.9)$$

and the supremum is achieved for some  $w \in \mathcal{W}$  that can be constructed from the previous lemmas. Since  $\rho(WL_c) \geq 1$  iff  $\exists x_0 \neq 0$  such that the right-hand side of

(15.9) is  $\geq 0$ , we have, by (15.9), that  $\rho(WL_c) \geq 1$  iff  $\exists w \in \mathcal{W}$ ,  $w \neq 0$  such that  $\|P_+z\|_2^2 \geq \|w\|_2^2$ . But this is true iff  $\sup_{w \in \mathcal{B}\mathcal{W}} \|\Gamma w\|_2 \geq 1$ .

If  $L_c$  is not invertible, we need only restrict  $x_0$  in (15.9) to  $\text{Im}(L_c)$ , and then the above argument generalizes in a straightforward way.  $\square$

In the differential game problem considered later and in the  $\mathcal{H}_\infty$  optimal control problem, we will make use of the adjoint  $\Gamma^* : \mathcal{H}_2 \rightarrow \mathcal{W}$ , which is given by

$$\Gamma^*z = \begin{bmatrix} P_-(G_1^\sim z) \\ G_2^\sim z \end{bmatrix} = \begin{bmatrix} P_-G_1^\sim \\ G_2^\sim \end{bmatrix} z \quad (15.10)$$

where  $P_-Gz := P_-(Gz) = (P_-M_G)z$ . That the expression in (15.10) is actually the adjoint of  $\Gamma$  is easily verified from the definition of the inner product on vector-valued  $\mathcal{L}_2$  space. The adjoint of  $\Gamma : \mathcal{W} \rightarrow \mathcal{H}_2$  is the operator  $\Gamma^* : \mathcal{H}_2 \rightarrow \mathcal{W}$  such that  $\langle z, \Gamma w \rangle = \langle \Gamma^*z, w \rangle$  for all  $w \in \mathcal{W}$ ,  $z \in \mathcal{H}_2$ . By definition, we have

$$\begin{aligned} \langle z, \Gamma w \rangle &= \langle z, P_+(G_1w_1 + G_2w_2) \rangle = \langle z, G_1w_1 \rangle + \langle z, G_2w_2 \rangle \\ &= \langle P_-(G_1^\sim z), w_1 \rangle + \langle G_2^\sim z, w_2 \rangle \\ &= \langle \Gamma^*z, w \rangle. \end{aligned}$$

The mixed Hankel-Toeplitz operator just studied is the so-called  $2 \times 1$ -block mixed Hankel-Toeplitz operator. There is a  $2 \times 2$ -block generalization.

## 15.4 Mixed Hankel-Toeplitz Operators: The General Case\*

Historically, the mixed Hankel-Toeplitz operators have played important roles in  $\mathcal{H}_\infty$  theory, so it is interesting to consider the  $2 \times 2$ -block generalization of Lemma 15.6. In fact, the whole  $\mathcal{H}_\infty$  control theory can be developed using these tools. See Section 17.7 in Chapter 17. The proof of Lemma 15.7 below is completely straightforward and fairly short, given the other results in the previous sections. Suppose that

$$G(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} = \left[ \begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right] =: \left[ \begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] \in \mathcal{RH}_\infty.$$

Denote

$$\begin{aligned} D_{\bullet 2} &:= \begin{bmatrix} D_{12} \\ D_{22} \end{bmatrix} & D_{2\bullet} &:= \begin{bmatrix} D_{21} & D_{22} \end{bmatrix} \\ R_x &:= I - D_{\bullet 2}^* D_{\bullet 2} & R_y &:= I - D_{2\bullet} D_{2\bullet}^* \end{aligned}$$

$$H_X := \begin{bmatrix} A & 0 \\ -C^*C & -A^* \end{bmatrix} + \begin{bmatrix} B_2 \\ -C^*D_{\bullet 2} \end{bmatrix} R_x^{-1} \begin{bmatrix} D_{\bullet 2}^*C & B_2^* \end{bmatrix}$$

$$H_Y := \begin{bmatrix} A^* & 0 \\ -BB^* & -A \end{bmatrix} + \begin{bmatrix} C_2^* \\ -BD_{2\bullet}^* \end{bmatrix} R_y^{-1} \begin{bmatrix} D_{2\bullet}B^* & C_2 \end{bmatrix}.$$

Define  $\mathcal{W} = \mathcal{H}_2^\perp \oplus \mathcal{L}_2$ ,  $\mathcal{Z} = \mathcal{H}_2 \oplus \mathcal{L}_2$ , and  $\Gamma : \mathcal{W} \rightarrow \mathcal{Z}$  as

$$\Gamma \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} P_+ & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}.$$

**Lemma 15.7**  $\sup_{w \in \mathcal{BW}} \|\Gamma w\|_2 < 1$  holds iff the following three conditions hold:

- (i)  $\bar{\sigma}(D_{\bullet 2}) < 1$  and  $H_X \in \text{dom}(\text{Ric})$ ;
- (ii)  $\bar{\sigma}(D_{2\bullet}) < 1$  and  $H_Y \in \text{dom}(\text{Ric})$ ;
- (iii)  $\rho(XY) < 1$  for  $X = \text{Ric}(H_X)$  and  $Y = \text{Ric}(H_Y)$ .

**Proof.** The mixed Hankel-Toeplitz operator can be written as

$$\Gamma \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = P_+ G \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} + \begin{bmatrix} 0 \\ P_- \begin{bmatrix} G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \end{bmatrix}.$$

Hence

$$\|\Gamma w\|_2^2 = \|P_+ G w\|_2^2 + \|P_- \begin{bmatrix} G_{21} & G_{22} \end{bmatrix} w\|_2^2.$$

So  $\sup_{w \in \mathcal{BW}} \|\Gamma w\| < 1$  implies

$$\sup_{w \in \mathcal{BW}} \|P_+ G w\| < 1$$

and

$$\sup_{w \in \mathcal{BW}} \|P_- \begin{bmatrix} G_{21} & G_{22} \end{bmatrix} w\| < 1.$$

But

$$\left\| \begin{bmatrix} G_{12} \\ G_{22} \end{bmatrix} \right\|_\infty = \sup_{w_2 \in \mathcal{H}_2} \left\| P_+ G \begin{bmatrix} 0 \\ w_2 \end{bmatrix} \right\|_2 \leq \sup_{w \in \mathcal{BW}} \|P_+ G w\| < 1$$

and

$$\left\| \begin{bmatrix} G_{21} & G_{22} \end{bmatrix} \right\|_\infty = \sup_{w \in \mathcal{H}_2^\perp} \left\| P_- \begin{bmatrix} G_{21} & G_{22} \end{bmatrix} w \right\|_2 \leq \sup_{w \in \mathcal{BW}} \|P_- \begin{bmatrix} G_{21} & G_{22} \end{bmatrix} w\| < 1.$$

These two inequalities then imply that (i) and (ii) are necessary. Analogous to the proof of Lemma 15.6, we will show that given conditions (i) and (ii),  $\sup_{w \in \mathcal{BW}} \|\Gamma w\|_2 < 1$



holds iff condition (iii) holds. We will do this by showing, equivalently, that  $\rho(XY) \geq 1$  iff  $\sup_{w \in \mathcal{B}\mathcal{W}} \|\Gamma w\|_2 \geq 1$ . By definition of  $\mathcal{W}$ , if  $w \in \mathcal{W}$  then

$$\|\Gamma w\|_2^2 - \|w\|_2^2 = (\|P_+ z\|_2^2 - \|P_+ w_2\|_2^2) + \left( \|P_- \begin{bmatrix} G_{21} & G_{22} \end{bmatrix} w\|_2^2 - \|P_- w\|_2^2 \right).$$

Thus if  $Y$  is invertible, then Lemma 15.3 and 15.5 yield

$$\sup_{w \in \mathcal{W}} \{ \|\Gamma w\|_2^2 - \|w\|_2^2 \mid x(0) = x_0 \} = x_0^* X x_0 - x_0^* Y^{-1} x_0.$$

Now the same arguments as in the proof of Lemma 15.6 give the desired conclusion.  $\square$

## 15.5 Linear Quadratic Max-Min Problem

Consider the dynamical system

$$\dot{x} = Ax + B_1 w + B_2 u \quad (15.11)$$

$$z = C_1 x + D_{12} u \quad (15.12)$$

with the following assumptions:

(i)  $(C_1, A)$  is observable;

(ii)  $(A, B_2)$  is stabilizable;

(iii)  $D_{12}^* \begin{bmatrix} C_1 & D_{12} \end{bmatrix} = \begin{bmatrix} 0 & I \end{bmatrix}$ .

In this section, we are interested in answering the following question: when

$$\sup_{w \in \mathcal{B}\mathcal{L}_{2+}} \min_{u \in \mathcal{L}_{2+}} \|z\|_2 < 1?$$

**Remark 15.3** This max-min problem is a game problem in the sense that  $u$  is chosen to minimize the quadratic norm of  $z$  and that  $w$  is chosen to maximize the norm. In other words, inputs  $u$  and  $w$  act as “opposing players”. This linear quadratic max-min problem can be reformulated in the traditional fashion as in the previous sections:

$$\sup_{w \in \mathcal{L}_{2+}} \min_{u \in \mathcal{L}_{2+}} \int_0^\infty (\|z\|^2 - \|w\|^2) dt = \sup_{w \in \mathcal{L}_{2+}} \min_{u \in \mathcal{L}_{2+}} (\|z\|_2^2 - \|w\|_2^2).$$

A conventional game problem setup would be to consider the min-max problem, i.e., switching the order of sup and min. However, it will be seen that they are equivalent and that a saddle point exists. By saying that, we would like to warn readers that this may not be true in the general case where  $z = C_1 x + D_{11} w + D_{12} u$  and  $D_{11} \neq 0$ . In that case, it is possible that  $\sup_w \inf_u < \inf_u \sup_w$ . This will be elaborated in Chapter 17.

It should also be pointed out that the results presented here still hold, subject to some minor modifications, if the assumptions (i) and (iii) on the dynamical system are relaxed to:

(i)',  $\begin{bmatrix} A - j\omega I & B_2 \\ C_1 & D_{12} \end{bmatrix}$  has full column rank for all  $\omega \in \mathbb{R}$ , and

(iii)'  $D_{12}$  has full column rank. ♡

It is clear from the assumptions that  $H_2 \in \text{dom}(\text{Ric})$  and  $X_2 = \text{Ric}(H_2) > 0$ , where

$$H_2 = \begin{bmatrix} A & -B_2 B_2^* \\ -C_1^* C_1 & -A^* \end{bmatrix}.$$

Let  $F_2 = -B_2^* X_2$  and  $D_\perp$  be such that  $[D_{12} \ D_\perp]$  is an orthogonal matrix. Define

$$G_c(s) := \left[ \begin{array}{c|c} A_{F_2} & I \\ \hline C_{1F_2} & 0 \end{array} \right], \quad U(s) := \left[ \begin{array}{c|c} A_{F_2} & B_2 \\ \hline C_{1F_2} & D_{12} \end{array} \right]$$

and

$$U_\perp = \left[ \begin{array}{c|c} A_{F_2} & -X_2^{-1} C_1^* D_\perp \\ \hline C_{1F_2} & D_\perp \end{array} \right] \quad (15.13)$$

where  $A_{F_2} = A + B_2 F_2$  and  $C_{1F_2} = C_1 + D_{12} F_2$ . The following is easily proven using Lemma 13.29 by obtaining a state-space realization and by eliminating uncontrollable states using a little algebra involving the Riccati equation for  $X_2$ .

**Lemma 15.8**  $[U \ U_\perp]$  is square and inner and a realization for  $G_c^\sim \begin{bmatrix} U & U_\perp \end{bmatrix}$  is

$$G_c^\sim \begin{bmatrix} U & U_\perp \end{bmatrix} = \left[ \begin{array}{cc|c} A_{F_2} & -B_2 & X_2^{-1} C_1^* D_\perp \\ \hline X_2 & 0 & 0 \end{array} \right] \in \mathcal{RH}_2. \quad (15.14)$$

This implies that  $U$  and  $U_\perp$  are each inner and that both  $U_\perp^\sim G_c$  and  $U^\sim G_c$  are in  $\mathcal{RH}_2^\perp$ . To answer our earlier question, define a Hamiltonian matrix  $H_\infty$  and the associated Riccati equation as

$$H_\infty := \begin{bmatrix} A & B_1 B_1^* - B_2 B_2^* \\ -C_1^* C_1 & -A^* \end{bmatrix}$$

$$A^* X_\infty + X_\infty A + X_\infty B_1 B_1^* X_\infty - X_\infty B_2 B_2^* X_\infty + C_1^* C_1 = 0.$$

**Theorem 15.9**  $\sup_{w \in B\mathcal{L}_{2+}} \min_{u \in \mathcal{L}_{2+}} \|z\|_2 < 1$  if and only if  $H_\infty \in \text{dom}(\text{Ric})$  and  $X_\infty = \text{Ric}(H_\infty) > 0$ . Furthermore, if the condition is satisfied, then  $u = F_\infty x$  with  $F_\infty := -B_2^* X_\infty$  is an optimal control.

**Proof.** ( $\Rightarrow$ ) As in Chapter 14, define  $\nu := u - F_2 x$  to get

$$z = G_c B_1 w + U \nu.$$

Then the hypothesis implies that

$$\sup_{w \in \mathcal{B}\mathcal{H}_2} \min_{\nu \in \mathcal{H}_2} \|z\|_2 < 1. \quad (15.15)$$

Since by Lemma 15.8  $[U \ U_\perp]$  is square and inner,  $\|z\|_2 = \|[U \ U_\perp]^\sim z\|_2$ , and

$$\begin{bmatrix} U & U_\perp \end{bmatrix}^\sim z = \begin{bmatrix} U^\sim G_c B_1 w + \nu \\ U_\perp^\sim G_c B_1 w \end{bmatrix} = \begin{bmatrix} P_- (U^\sim G_c B_1 w) + P_+ (U^\sim G_c B_1 w + \nu) \\ U_\perp^\sim G_c B_1 w \end{bmatrix}.$$

Thus

$$\sup_{w \in \mathcal{B}\mathcal{H}_2} \min_{\nu \in \mathcal{H}_2} \|z\|_2 = \sup_{w \in \mathcal{B}\mathcal{H}_2} \min_{\nu \in \mathcal{H}_2} \left\| \begin{bmatrix} P_- (U^\sim G_c B_1 w) + P_+ (U^\sim G_c B_1 w + \nu) \\ U_\perp^\sim G_c B_1 w \end{bmatrix} \right\|_2,$$

and the right hand of the above equation is minimized by  $\nu = -P_+ (U^\sim G_c B_1 w)$ ; we have

$$\begin{aligned} \sup_{w \in \mathcal{B}\mathcal{H}_2} \min_{\nu \in \mathcal{H}_2} \|z\|_2 &= \sup_{w \in \mathcal{B}\mathcal{H}_2} \left\| \begin{bmatrix} P_- (U^\sim G_c B_1 w) \\ U_\perp^\sim G_c B_1 w \end{bmatrix} \right\|_2 \\ &=: \sup_{w \in \mathcal{B}\mathcal{H}_2} \|\Gamma^* w\|_2 < 1 \end{aligned}$$

where  $\Gamma^* : \mathcal{L}_{2+} \rightarrow \mathcal{W}$  is defined as

$$\Gamma^* w = \begin{bmatrix} P_- (U^\sim G_c B_1 w) \\ U_\perp^\sim G_c B_1 w \end{bmatrix} = \begin{bmatrix} P_- U^\sim \\ U_\perp^\sim \end{bmatrix} G_c B_1 w$$

with

$$\mathcal{W} := \left\{ \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \middle| q_1 \in \mathcal{H}_2^\perp, q_2 \in \mathcal{L}_2 \right\}.$$

Note that from equation (15.10) the adjoint operator  $\Gamma : \mathcal{W} \rightarrow \mathcal{H}_2$  is given by

$$\Gamma \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} = P_+ (B_1^* G_c^\sim (U q_1 + U_\perp q_2)) = P_+ B_1^* G_c^\sim \begin{bmatrix} U & U_\perp \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$$

where  $G_c^\sim [U \ U_\perp] \in \mathcal{R}\mathcal{H}_2$  has the realization in (15.14). So we have

$$\sup_{q \in \mathcal{B}\mathcal{W}} \|\Gamma q\|_2 < 1.$$

This is just the condition (15.8), so from Lemma 15.6 and equation (15.14) we have that

$$H_W := \begin{bmatrix} A_{F_2} & X_2^{-1} C_1^* C_1 X_2^{-1} \\ -X_2 B_1 B_1^* X_2 & -A_{F_2}^* \end{bmatrix} \in \text{dom}(\text{Ric})$$

and  $W = \text{Ric}(H_W) \geq 0$ . Note that the observability of  $(C_1, A)$  implies  $X_2 > 0$ . Furthermore, the controllability Gramian for the system (15.14) is  $X_2^{-1}$  since

$$A_{F_2} X_2^{-1} + X_2^{-1} A_{F_2}^* + B_2 B_2^* + X_2^{-1} C_1^* C_1 X_2^{-1} = 0.$$

Lemma 15.6 also implies  $\rho(W X_2^{-1}) < 1$  or, equivalently,  $X_2 > W$ . Using the Riccati equation for  $X_2$ , one can verify that  $T := \begin{bmatrix} -I & X_2^{-1} \\ -X_2 & 0 \end{bmatrix}$  provides a similarity transformation between  $H_\infty$  and  $H_W$ , i.e.,  $H_\infty = T H_W T^{-1}$ . Then

$$\mathcal{X}_-(H_\infty) = T \mathcal{X}_-(H_W) = T \text{Im} \begin{bmatrix} I \\ W \end{bmatrix} = \text{Im} \begin{bmatrix} I - X_2^{-1} W \\ X_2 \end{bmatrix},$$

so  $H_\infty \in \text{dom}(\text{Ric})$  and  $X_\infty = X_2(X_2 - W)^{-1} X_2 > 0$ .

( $\Leftarrow$ ) If  $H_\infty \in \text{dom}(\text{Ric})$  and  $X_\infty = \text{Ric}(H_\infty) > 0$ ,  $A + B_1 B_1^* X_\infty - B_2 B_2^* X_\infty$  is stable. Define

$$A_{F_\infty} := A + B_2 F_\infty, \quad C_{1F_\infty} := C_1 + D_{12} F_\infty.$$

Then the Riccati equation can be written as

$$A_{F_\infty}^* X_\infty + X_\infty A_{F_\infty} + C_{1F_\infty}^* C_{1F_\infty} + X_\infty B_1 B_1^* X_\infty = 0.$$

We conclude from Lyapunov theory that  $A_{F_\infty}$  is stable since  $(A_{F_\infty}, B_1^* X_\infty)$  is detectable and  $X_\infty > 0$ . Now with the given control law  $u = F_\infty x$ , the dynamical system becomes

$$\begin{aligned} \dot{x} &= A_{F_\infty} x + B_1 w \\ z &= C_{1F_\infty} x. \end{aligned}$$

So by Corollary 13.24,  $\|T_{zw}\|_\infty < 1$ , i.e.,  $\sup_{w \in \mathcal{L}_{2+}} \|z\|_2 < 1$  for the given control law.  $\square$

Theorem 15.9 will be used in the next chapter to solve the FI  $\mathcal{H}_\infty$  control problem.

## 15.6 Notes and References

Differential game is a well-studied topic, see e.g., Bryson and Ho [1975]. The paper by Mageirou and Ho [1977] is one of the early papers that are relevant to the topics covered in this chapter. The current setup and proof are taken from Doyle, Glover, Khargonekar, and Francis [1989]. The application of game theoretic results to  $\mathcal{H}_\infty$  problems can be found in Başar and Bernhard [1991], Khargonekar, Petersen, and Zhou [1990], Limebeer, Anderson, Khargonekar, and Green [1992], and references therein.